CITY COLLEGE

Approved by AICTE, New Delhi. Recognized by Govt. of Karnataka. Affiliated to Bangalore University, accredited by NAAC with "B+" Grade No. 27/2, 33rd Cross, 2nd Main Rd, 7th Block, Jayanagar, Bengaluru, Karnataka 560070 Phone: 6364914155 / 56, +91 80 26649277
Website: www.citycollege.ac.in Email: admin@citycollege.ac.in

Project Report On

"Aadhaar-Related phising and identity theft cases"

Submitted in partial fulfilment of the requirement for the project work for the degree of

BACHELOUR OF COMPUTER APPLICATION (BCA)
2025–26

BANGALORE UNIVERSITY

Under the Guidance Of
Ms. Chandana Khatavakar
Assistant Professor
Department of Computer Application

Submitted by
MANASA N
(U03BX23S0046)

CITY COLLEGE, JAYANAGAR, BENGALURU
DEPARTMENT OF COMPUTER APPLICATION

CERTIFICATE

**This is to certify that the project synopsis entitled is" Aadhaar-Related phising and identity theft cases"
submitted by Ms.MANASA N (USN: U03BX23S0046) in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from Bangalore University for the academic year 2025–2026.
The project synopsis has been prepared under the guidance and supervision of the undersigned. The synopsis is submitted for academic evaluation in accordance with the applicable department and university guidelines.**

Project Guide **Ms. Chandana Khatavakar** **Assistant Professor**	**Head of the Department** **Dr. Nageshwari** **Department of Computer Application**	**Principal** **City College Jayanagar, Bengaluru**

Place: Bengaluru
Date: 10-10-2026

TABLE OF CONTENTS

Sl. No.	Contents
1.	Introduction
2.	Incident Description
3.	Attack Type / Vulnerability
4.	Root Cause Analysis
5.	Impact Analysis
6.	Detection and Response
7.	Security Measures and Mitigation
8.	Lessons Learned
9.	Recommendations
10.	Conclusion
11.	References

Aadhaar Phishing and Identity Theft

Case Study: The October 2023 ICMR Aadhaar and Passport Data Leak

Author: Manasa N
Date: 08 October 2026
Category: Cybersecurity Incident Case Study

Disclaimer. *This case study is prepared for educational and awareness purposes only. It is based on publicly available news reports, parliamentary records and expert commentary, and contains no confidential information. Allegations reported in the media remain allegations unless established by a court. The author is not affiliated with UIDAI, ICMR, CERT-In, Resecurity or any government body, and nothing here is legal or financial advice. No personal data of any individual is reproduced. Readers should confirm current guidance at uidai.gov.in.*

Abstract

Aadhaar, India's 12-digit biometric identity number, underpins banking, telecom, welfare and tax services, which makes leaked or phished Aadhaar details a high-value asset for fraudsters. This case study examines the October 2023 incident in which a threat actor advertised 815 million 'Indian Citizen Aadhaar and Passport' records on a dark web forum, reportedly drawn from COVID-19 test records linked to the Indian Council of Medical Research. The listing was first reported by the cyber intelligence firm Resecurity, verified through CERT-In, and followed by four arrests by Delhi Police in December 2023. The study traces the timeline, explains how exposed identity data feeds phishing, vishing and identity theft, analyses likely root causes using only publicly reported facts, and assesses the impact on individuals and institutions. It then reviews detection and response, safeguards such as biometric lock, masked Aadhaar and Virtual ID, and lessons for organisations that hold identity data. It closes with practical recommendations for citizens, institutions and policymakers. All facts come from public reporting and parliamentary records.

Keywords: Aadhaar, phishing, identity theft, data breach, ICMR, CERT-In, biometric lock

1. Introduction

Aadhaar is a 12-digit identity number issued by the Unique Identification Authority of India (UIDAI) and linked to demographic data and biometrics. Over the past decade it has become the connecting key for bank accounts, mobile SIM issuance, subsidies, tax filings and digital know-your-customer (KYC) checks. That convenience concentrates risk. Anyone holding a person's Aadhaar number, phone number and address can attempt to impersonate that person, or craft a phishing message that appears to come from a trusted authority.

This case study examines the October 2023 exposure of records described as 'Indian Citizen Aadhaar and Passport' data, reportedly linked to COVID-19 testing records associated with the Indian Council of Medical Research (ICMR) [1][2]. It uses the incident to show how leaked identity data and Aadhaar-themed phishing reinforce each other in a single fraud pipeline. The aim is educational: to document what is publicly known, separate verified facts from allegations, and draw lessons for any organisation that handles national identity data.

2. Incident Description

On 9 October 2023, a threat actor using the alias pwn0001 posted a thread on Breach Forums offering access to 815 million records labelled as Indian citizen Aadhaar and passport data [1][2]. The US-based firm Resecurity noticed the listing and reported that the seller shared a sample of about 100,000 records. Its analysts confirmed that Aadhaar numbers in the sample were valid by checking them against a government 'Verify Aadhaar' feature [1]. The seller reportedly asked US$80,000 for the full dataset and declined to explain how it was obtained [1]. The records were said to include names, phone numbers, and temporary and permanent addresses alongside Aadhaar and passport details, and the seller claimed they came from COVID-19 test records held by ICMR [2][9].

CERT-In informed ICMR and verified the sample with the departments concerned. Reportedly, 50 individuals picked from the sample of roughly one lakh records were found to match actual data [4]. Because COVID-19 test data sits across several government bodies, including NIC, ICMR and the health ministry, pinpointing the source system was difficult [9]. Delhi Police registered an FIR and, in December 2023, arrested four men from Odisha, Haryana and Jhansi, who reportedly met on a gaming platform and wanted quick money [4]. They were later booked in a separate data-brokering case [5].

Table 1. Timeline of key events

Date	Event
Jan 2018	The Tribune reports buying access to Aadhaar details from an anonymous WhatsApp seller; UIDAI says its data is secure [8]
30 Aug 2023	An actor named Lucius reportedly claims a large data leak [3]
9 Oct 2023	pwn0001 lists 815 million records on Breach Forums [1][2]
31 Oct 2023	Resecurity findings widely reported; CBI probe expected after an ICMR complaint [1][2]
8 Dec 2023	Government tells Rajya Sabha that CERT-In tracked 165 citizen data breach incidents from Jan 2018 to Oct 2023, with no breach of UIDAI's CIDR [6]
Dec 2023	Delhi Police arrest four suspects [4][10]
Late Dec 2023	The four are booked in a separate data-brokering case [5]

3.Attack Type / Vulnerability

The core incident is a data breach followed by data brokering on a cybercrime forum, a form of large-scale personal data exfiltration. Phishing and identity theft are the downstream attack types this data enables. Public reporting does not state the initial intrusion method, so no specific software vulnerability is attributed here.

Three patterns connect leaked Aadhaar data with fraud. First, targeted phishing and vishing: a fraudster who knows a victim's name, phone number and address can claim that Aadhaar is suspended or KYC is incomplete, then push the victim to a lookalike site that harvests the Aadhaar number, OTP and bank details [11]. Second, identity theft: the same data can support fake bank accounts, loans or SIM cards opened in a victim's name [12]. Third, Aadhaar-enabled Payment System (AePS) fraud: where fingerprints are cloned or captured, criminals can withdraw money using an Aadhaar number, which is why UIDAI promotes the biometric lock [11].

The underlying weakness is structural rather than a single bug. Aadhaar numbers are collected and copied as photocopies, forms and databases outside UIDAI's control, and the number has long been treated as freely shareable. Each copy is a potential leak point that the central system cannot protect.

4.Root Cause Analysis

The technical root cause has not been publicly established. Officials were reported to be still questioning the suspects about how the data was stolen [4]. The analysis below therefore separates what is reported from what is plausible.

- **Fragmented custody.** COVID-19 test records were spread across NIC, ICMR and the health ministry, so no single owner could quickly confirm which system leaked [9].
- **Sustained hostile attention.** Reports cited thousands of attempts to hack the ICMR website in 2022, indicating a known threat environment [2].
- **Recycled data.** Commentary on the 2018 Tribune episode suggested the data came from previously breached databases, showing how exposed data circulates and is repackaged [8].
- **Low barrier to monetisation.** The suspects were described as a B.Tech graduate and school dropouts who met on a gaming platform, which shows that once data is exposed, small groups can trade it [4].

Plausible but unconfirmed factors common in such incidents include weak access control on a data-sharing interface, stolen credentials or insider access, and retention of pandemic-era records long after their purpose ended. These are hypotheses, not findings.

5.Impact Analysis

People whose records were exposed face higher risk of targeted phishing, SIM fraud, loan and KYC fraud, and harassment. Phone numbers and addresses paired with genuine Aadhaar numbers make scam messages far more convincing. Unlike a password, an Aadhaar number cannot be changed, so exposure is effectively permanent. Public reports have not quantified financial losses from this specific dataset, and this study does not claim a figure.

For institutions, the episode brought scrutiny, police and CBI involvement, and loss of trust. It also exposed a communication gap: the government stated in Parliament that no breach occurred from UIDAI's CIDR [6][7], while the leaked sample contained valid Aadhaar numbers [1]. Both can be true because data can leak from downstream holders, but citizens cannot easily tell the difference.

Table 2. Impact by stakeholder

Stakeholder	Impact	Severity
Citizens	Targeted phishing, identity misuse, permanent exposure of an unchangeable identifier	High
Banks and telecom	Higher fraud-handling and KYC verification costs	Medium
ICMR and health bodies	Investigation burden, reputational damage, scrutiny of data governance	High
UIDAI and government	Pressure to clarify what was and was not breached; trust in digital public infrastructure	Medium

6.Detection and Response

Detection came from outside the data owner. Resecurity's monitoring of underground forums spotted the listing, and its findings were widely reported on 31 October 2023 [1][2]. CERT-In alerted ICMR and authenticated the sample by cross-checking with the departments concerned [4][9]. Delhi Police took suo motu cognisance, registered an FIR and arrested four suspects found advertising the data for sale. Charges cited Section 420 of the IPC (cheating) and Section 43 of the IT Act [4][10]. The CBI was also reported to be examining the matter [10].

Strengths: the sample was verified quickly and arrests followed about ten weeks after the listing. Weaknesses: the data owner did not detect the exfiltration itself, the source system stayed unclear, and the public explanation of how the data left remained thin.

7.Security Measures and Mitigation

Table 3. Controls that reduce phishing and identity theft risk

Control	Purpose	Who acts
Biometric lock	Blocks fingerprint and iris authentication until the holder unlocks it, which blunts AePS fraud [11][13]	Citizen
Masked Aadhaar	Shows only the last four digits when a copy is required [12]	Citizen
Virtual ID (VID)	Temporary number used in place of the Aadhaar number for authentication [11]	Citizen

Official channels only	Update details only through uidai.gov.in or an Aadhaar Seva Kendra; ignore update links sent by SMS or WhatsApp [11]	Citizen
Data minimisation and retention limits	Collect only what is needed and delete records when their purpose ends	Institution
Encryption, access logs, dark-web monitoring	Limit exposure, detect misuse early, and notify CERT-In promptly [6]	Institution

8.Lessons Learned

- Identity data is only as safe as its weakest copy. Protecting a central database does not protect forms, spreadsheets and downstream systems.
- Organisations should detect their own leaks. Relying on outside researchers delays response.
- Clear data ownership speeds up incident handling; shared custody slowed source identification here.
- Retention beyond purpose multiplies risk, especially for emergency datasets such as pandemic records.
- Precise public communication, stating exactly what was and was not affected, protects trust better than blanket denials.

9.Recommendations

Citizens. Lock biometrics, use masked Aadhaar or VID wherever possible, and never share OTPs. Treat any message about Aadhaar suspension or expiry as suspicious, and open official sites by typing the address yourself. Report suspected fraud quickly through the official helplines and cybercrime portal listed on uidai.gov.in [13].

Organisations. Map where identity data lives, minimise and expire it, enforce least-privilege access, log every bulk export, and monitor underground forums for your own data. Run phishing-awareness training and rehearse breach response with defined owners.

Policymakers. Set uniform retention and breach-disclosure rules for all holders of identity data, require prompt public statements that distinguish central-system integrity from downstream leaks, and fund citizen awareness programmes.

10.Conclusion

The October 2023 listing shows that the largest risk to Aadhaar holders may lie less in the central repository than in the many downstream systems that copy identity data. Once such data is on sale, it fuels phishing, identity theft and payment fraud for years. Fast verification by CERT-In and the arrests by Delhi Police show that response capacity exists, but prevention depends on data minimisation, clear ownership, self-detection and informed citizens. Biometric lock, masked Aadhaar and Virtual ID give individuals real protection today, while institutions must close the gaps that allow data to leave in the first place.

Figure 1. Attack chain and response flow

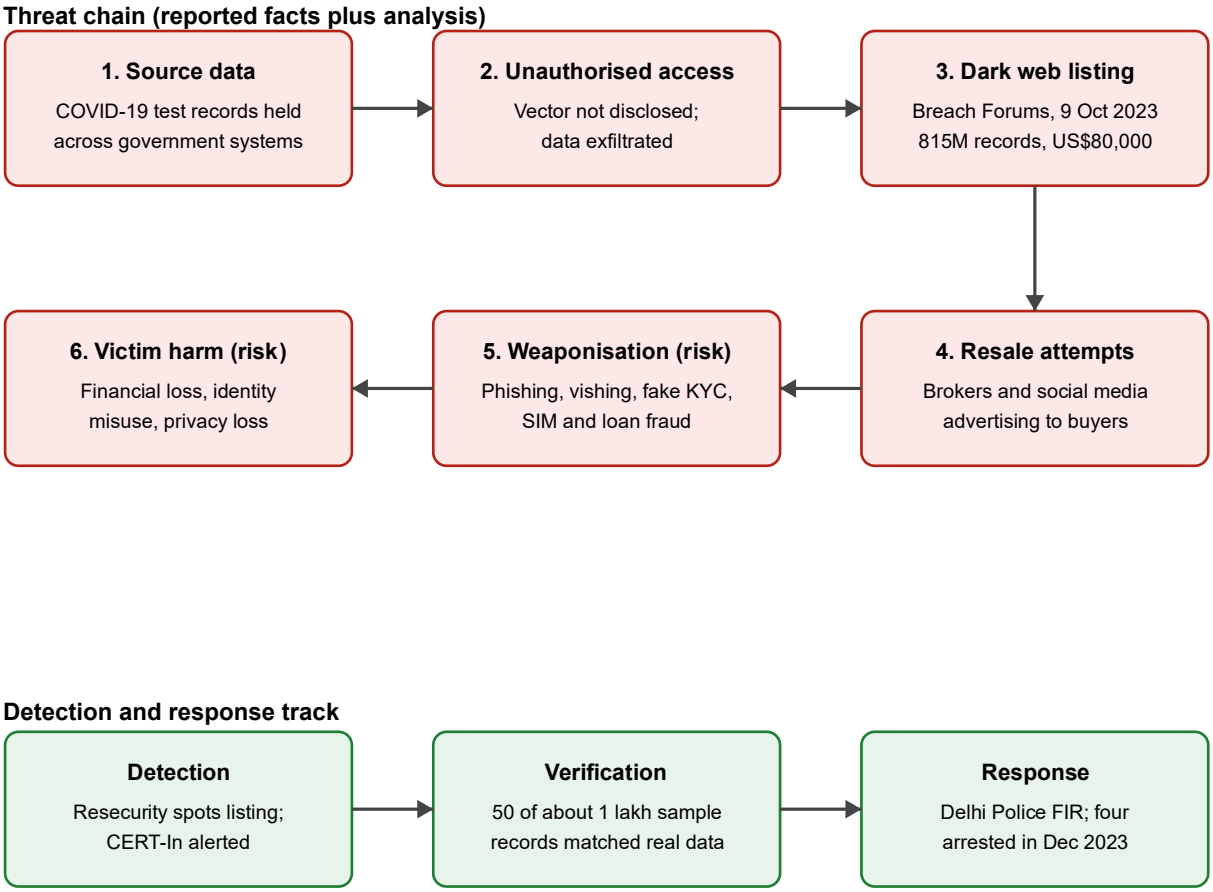


Figure 1. How exposed Aadhaar data moves from source to fraud, and how detection and response followed. Steps 5 and 6 are risk analysis, not reported victim counts. Source: author's analysis based on references [1]-[5], [9]-[12].

11. References

[1] IANS via Moneylife, 'ICMR Data Leak Reveals Personal Info of 81.5 Cr Indians: Report', 31 Oct 2023. https://moneylife.in/article/icmr-data-leak-reveals-personal-info-of-815-cr-indians-report/72439.html

[2] Deccan Herald, 'Data of 81.5 crore citizens up for sale in biggest data breach in India: Report', 31 Oct2023. https://www.deccanherald.com/amp/story/india%2Fdata-of-815-crore-citizens-up-for-sale-inbiggest-data-breach-in-india-report-2749794

[3] Business Today, 'Data Breach of 81.5 crore Indians: Hacker allegedly leaks Aadhaar, passport,personal details on dark web', 31 Oct 2023. https://www.businesstoday.in/amp/technology/news/st ory/data-breach-of-815-crore-indians-hacker-allegedly-leaks-aadhaar-passport-personal-details-ondark-web-403903-2023-10-31

[4] Maktoob Media (citing The Indian Express), 'Delhi Police arrests four persons in ICMR data leakcase', Dec 2023. https://maktoobmedia.com/india/delhi-police-arrests-four-persons-in-icmr-data-leak-case/

[5] DataBreaches.Net (citing ThePrint), '4 arrested in ICMR data leak case now booked in separateincident of data brokering', 28 Dec 2023. https://www.databreaches.net/4-arrested-in-indian-council -of-medical-research-data-leak-case-now-booked-in-separate-incident-of-data-brokering/

[6] Rajya Sabha Unstarred Question No. 737, Ministry of Electronics and IT, answered 8 Dec 2023. https://rsdebate.nic.in/bitstream/123456789/746472/1/PQ_262_08122023_U737_p339_p339.pdf

[7] Rajya Sabha Unstarred Question No. 878, Ministry of Electronics and IT, answered 9 Feb 2024. https://rsdebate.nic.in/bitstream/123456789/748121/1/PQ_263_09022024_U878_p308_p309.pdf

[8] Scroll.in, '165 breaches of data of Indians reported since 2018, says Centre', Dec 2023. https://scroll.in/latest/1060451/165-data-breaches-of-indians-reported-since-2018-says-centre

[9] BW Healthcare World, 'ICMR data breach: 815 Cr Indians data leaked on darkweb'.https://bwhealthcareworld.com/article/icmr-data-breach-815-cr-indians-data-leaked-on-darkweb-496963

[10] King Stubb and Kasiva, 'Arrests Made In The ICMR Data Breach Case', 2024. https://ksandk.com/newsletter/arrests-made-in-the-icmr-data-breach-case/

[11] Quick Heal Knowledge Centre, 'How to Prevent Aadhaar Card Scams'. https://www.quickheal.co.in/knowledge-centre/how-to-prevent-fraud-aadhar-card-scam/

[12] Poonawalla Fincorp, 'Aadhaar Card Fraud and Scams in India: Prevention Guide'.https://poonawallafincorp.com/blogs/financial-insights/aadhaar-card-fraud-scams-india-prevention-guide

[13] Unique Identification Authority of India (UIDAI), official website. https://uidai.gov.in

THANK YOU

DEPARTMENT OF BCA

CITY COLLEGE JAYNAGAR

CYBER SECURITY REPORT

Academic Year

2026-2027